

RURAL / VILLAGE LOCAL JOBS

SaaS Platform — High-Level Product Requirements & Project Plan

Updated edition: security, trust/bad-rating controls, and three separate dashboards included.

Item	Definition
Product	Multilingual rural/local jobs marketplace SaaS
Users	Worker, Job Provider, Admin
Focus	Village and rural practical, skilled, temporary, recurring, and permanent work
Core matching	Skills + location + availability + job preferences + reputation
Core automation	Job closes when filled; job expires after application deadline
Security principle	Role-based access, least privilege, secure authentication, validation, auditability

1. Product Vision

Create a mobile-first platform connecting rural/local workers with people and organizations that need nearby help. The platform is not a corporate/IT recruitment portal; it is a trusted local-work marketplace.

Primary value: a worker can find nearby work in a preferred language, while a job provider can quickly find available and reputable workers.

2. User Roles

Role	Main purpose
Worker	Offer skills, availability and experience; discover jobs; apply; communicate; work; receive and give ratings.
Job Provider	Post jobs; search workers; review applications; select/hire; communicate; complete jobs; rate workers.
Admin	Operate and secure the marketplace; moderate users/jobs; manage verification, ratings, reports, disputes, categories.

3. Three Separate Dashboards

The system must have three clearly separated dashboards with role-based access control. A user must never be able to access another role's dashboard simply by changing a URL or request.

A. Worker Dashboard

- Home / nearby recommended jobs.
- New matching job notifications.
- Search jobs by category, skill, distance, payment and date.
- My applications: pending, accepted, rejected, withdrawn.
- My active jobs and completed jobs.
- Availability calendar.
- Worker profile, skills, experience, rating and verification.
- Messages / job-specific chat.
- Call/contact option subject to privacy controls.
- Ratings and reviews received; ability to rate completed providers.
- Notification preferences and language settings.

B. Job Provider Dashboard

- Dashboard summary: open jobs, applications, hired workers, active work, completed jobs.
- Create and manage jobs.
- Search/filter workers by skills, distance, availability, experience and rating.
- View worker profiles and verification status.
- Review applications, shortlist and select workers.
- Job-specific chat/contact.
- Automatic FILLED/EXPIRED status visibility.
- Manage active/completed/cancelled/expired job history.
- Rate workers after completion and receive provider ratings.
- Notifications and language settings.

C. Admin Dashboard

- System overview: users, active jobs, filled jobs, expired jobs, reports, disputes and platform health.
- Worker management: verification, warnings, suspension, ban, restore.
- Job-provider management: verification, warnings, suspension, ban, restore.
- Job moderation and reported-content review.
- Ratings/reviews moderation and abuse detection.
- Reports, complaints and dispute resolution.
- Category, skill and multilingual content management.
- Notification templates and delivery monitoring.
- Security/audit logs.
- Analytics: registrations, jobs, applications, hires, completion, retention, geography.

4. Job Categories

Category	Examples
Household	House cleaner, house maid, cook, babysitter, gardener, washing/cleaning, elderly assistance
Agriculture	Farm worker, harvesting, planting, tractor operator, animal/cattle care
Construction	Mason, helper, painter, plumber, electrician, welder
Village/community	Gram road cleaning, drain cleaning, grass cutting, garbage collection, water-related work, event/setup workers
Local services	Carpenter, mechanic, tailor, barber, driver, delivery/helper

5. Multilingual & Rural-Friendly UX

- Launch target: Marathi + Hindi + English; expand based on usage.
- Localization architecture from day one for UI, categories, notifications, help and settings.
- Mobile-first responsive interface.
- Large touch targets, clear icons, simple forms and minimal typing.
- Low-bandwidth-conscious design.
- Future: voice-assisted job creation and search.

6. Worker Profile

- Name, profile photo, village/location and languages.

- Multiple controlled skills and experience.
- Expected hourly/day/month payment.
- Availability by date.
- Preferred categories and search radius.
- Completed jobs, rating/reviews and verification status.
- Privacy/contact preferences.

7. Job Posting

Field	Requirement
Title/category	Clear job title plus controlled category/skills
Location	Village/area plus coordinates where practical
Workers required	Number needed
Start	Date/time
Duration	Hourly, one day, multiple days, recurring or permanent
Payment	Amount + unit
Deadline	Application closing date/time
Requirements	Skills, experience, tools, conditions and other lawful requirements
Description	Detailed work expectations

8. Job Lifecycle & Automatic Rules

Recommended states: DRAFT OPEN FILLED / EXPIRED / CANCELLED IN_PROGRESS COMPLETED CLOSED, with DISPUTED available when necessary.

- FILLED: when selected/hired workers reach the required number. Close new applications and stop normal matching notifications.
- EXPIRED: when the application deadline passes while the job remains open. Remove from active listings and stop applications/normal notifications.
- Historical retention: do not permanently delete filled/expired/completed jobs solely because they leave active listings; retain records for history, ratings, disputes, analytics and audit.
- Server-side enforcement: these rules must be enforced in backend transactions/jobs, not only in frontend code.

9. Matching & Notifications

Matching signals: required skills, distance, availability, preferred category, payment fit, experience, rating/reputation and urgency.

Notification flow: new job matching engine eligible workers notification worker opens job applies.

- Notify only relevant workers; avoid sending every job to everyone.
- Worker can set radius, categories, minimum payment and notification preferences.
- Notify on new matching jobs, application status, selection, messages, reminders, cancellation and rating requests.
- Never send normal job-match notifications for FILLED, EXPIRED or CANCELLED jobs.

10. Ratings, Bad Ratings & Account Enforcement

Ratings should be tied to completed jobs. Both workers and job providers can rate each other. Reputation is a trust signal, but a single bad rating should not automatically ban an account.

Recommended trust policy

- Use overall rating plus dimensions such as work quality, punctuality, behavior and payment reliability.

- Detect repeated low ratings, suspicious rating patterns, fake reviews, retaliation, abuse and policy violations.
- Use thresholds and review history rather than a single complaint as the sole ban trigger.
- A very low reputation score can trigger reduced visibility, warning, verification review or temporary suspension.
- Serious safety/fraud/abuse violations can trigger immediate suspension while the case is investigated.
- Provide an appeal/review process where appropriate.
- The same trust framework applies to workers and job providers.

Suggested enforcement ladder

Level	Action
Normal	Account active; standard marketplace visibility
Warning	Notify user and explain the issue; continue monitoring
Restricted	Reduce visibility or restrict selected features while reviewing repeated issues
Suspended	Temporarily prevent job applications/posting and/or communication as appropriate
Banned	Permanently prevent access for confirmed serious/repeated violations, subject to policy/appeal rules

11. Security Architecture

Security must be treated as a first-class system requirement because the platform handles accounts, phone numbers, location, conversations, ratings, identity/verification information and potentially payments.

- Authentication: secure mobile OTP/passwordless or equivalent authentication; protect sessions/tokens; support logout and device/session management.
- Authorization: strict RBAC for Worker, Job Provider and Admin. Enforce permissions on every protected API.
- Least privilege: admin functions separated from normal user functions; sensitive operations require elevated permissions.
- Input validation: validate all request data server-side; never trust client-side values.
- API security: rate limiting, abuse protection, secure headers, request-size limits and safe error handling.
- Data protection: encrypt data in transit; encrypt sensitive data at rest where appropriate; protect secrets in a secret manager.
- Privacy: minimize collection, restrict phone/location exposure, define retention rules and provide appropriate account/data controls.
- File security: validate uploaded files, restrict types/sizes, isolate storage, scan where appropriate and use non-public object storage for sensitive documents.
- Chat safety: report/block controls, abuse monitoring and controlled access to conversations.
- Audit logs: record critical admin actions, bans, verification changes, job-status changes and other security events.
- Database security: parameterized queries/ORM protections, least-privileged DB users, backups and tested restore procedures.
- Operational security: separate development/staging/production, secure CI/CD, dependency updates, monitoring and incident response.

12. Security & Trust Scenarios

- User attempts to access another user's private profile data API denies access.
- Worker tries to apply after job is FILLED backend rejects application.
- Provider tries to accept more workers than required backend enforces capacity.
- Expired job receives an application backend rejects it.
- User repeatedly receives credible severe complaints account enters trust review/restriction.
- Admin bans user action is logged with actor, timestamp, reason and affected account.

- Suspicious rating activity ratings can be flagged for moderation.

13. Chat, Contact & Privacy

- Job-specific conversation between provider and worker.
- Real-time messaging can use WebSockets/realtime infrastructure.
- Call/contact option should respect privacy and consent; consider masked calling later.
- Message and notification events are tied to the relevant job.
- Report/block controls should be available.

14. Verification

- Mobile number verification as baseline.
- Basic profile/location verification.
- Optional identity verification.
- Skill/experience verification where practical.
- Clear meaning for verified badges.
- Admin review for suspicious or reported accounts.

15. Payments

For V1, consider recording agreed payment without processing money through the platform. This reduces financial, fraud, refund, payout and regulatory complexity during validation.

Later: gateway, payouts, refunds, commissions, dispute workflows, KYC/tax requirements and legally appropriate payment/escrow-like mechanisms.

16. Admin Moderation & Disputes

- Users can report jobs, workers or providers.
- Admin can review evidence and history.
- Admin can warn, restrict, suspend or ban.
- Admin decisions and reasons are audit logged.
- Dispute states: OPEN UNDER_REVIEW RESOLVED / REJECTED / ESCALATED.
- Preserve relevant job, message, application and rating history according to retention policy.

17. Technical High-Level Architecture

Layer	Recommended approach
Client	Next.js/React responsive mobile-first web app; native/mobile wrapper can be considered later
Backend	Java Spring Boot REST API
Database	PostgreSQL
Auth	Secure OTP/session or token-based architecture
Storage	Object storage for profile/document assets
Search	PostgreSQL search initially; dedicated search service if scale requires
Matching	Backend service/rules engine; later ML/AI recommendation layer
Realtime	WebSocket/realtime service for chat
Notifications	Push; SMS/WhatsApp integrations later
Maps	Geolocation/maps provider
Background jobs	Deadline expiration, notifications, reminders, moderation jobs
Observability	Central logs, metrics, alerts and audit logs

18. Core Data Model

Entity	Purpose
User	Common account/authentication data
WorkerProfile	Worker details
ProviderProfile	Job provider details
Skill / WorkerSkill	Controlled skills and worker relationships
Availability	Date/time availability
Job / JobSkill	Job and required skills
Application	Worker application
Assignment	Selected/hired worker relationship
Conversation / Message	Job-specific communication
Notification	Notification events/preferences
Review	Ratings/reviews tied to completed work
Verification	Verification status/evidence metadata
Report / Dispute	Trust, safety and complaints
Restriction / Ban	Warnings, suspensions, bans, reasons, timestamps and appeals
AuditLog	Critical admin/security actions
PaymentRecord	Future payment information

19. Three-Dashboard Permission Model

Capability	Worker	Job Provider	Admin
Manage own profile			
Search jobs		—	
Search workers	—		
Create jobs	—		
Apply to jobs		—	
Select/hire workers	—		
Chat			where authorized
Rate completed counterpart			moderation
Manage users	—	—	
Ban/suspend	—	—	
Manage categories/translations	—	—	
Security/audit tools	—	—	

20. MVP Scope

- Three role-based dashboards.
- Worker/provider registration and profiles.
- Marathi + Hindi + English localization foundation.
- Skills, categories, location and availability.
- Job creation and job search.

- Worker search for providers.
- Applications and selection.
- Automatic FILLED and EXPIRED rules.
- Matching job notifications.
- Job-specific chat/contact.
- Ratings/reviews for both sides.
- Basic verification and reporting.
- Admin moderation, warnings, suspension and ban.
- Security baseline: RBAC, validation, rate limiting, secure secrets, audit logs and backups.

21. Future Roadmap

V2	V3 / Scale
Worker teams	Advanced workforce management
SMS/WhatsApp workflows	AI-assisted matching/recommendations
Voice-assisted search/posting	Automated multilingual assistance
Stronger verification	Attendance and payroll workflows
Organization accounts	Marketplace payments and payouts
Recurring jobs	Advanced fraud/rating detection
Improved map experience	Regional/state-wide expansion
Advanced admin tools	Partner/API ecosystem

22. Development Roadmap

Phase	Deliverable
0 Discovery	Target geography, first categories, user rules, trust/safety policy, V1 scope
1 UX/UI	Three dashboards, mobile flows, multilingual UI, accessibility
2 Foundation	Auth, RBAC, PostgreSQL, localization, storage, environments
3 Worker	Profile, skills, availability, job search, applications
4 Provider	Job creation, worker search, applications, hiring
5 Marketplace	Matching, notifications, fill/expiry automation
6 Communication	Chat/contact
7 Trust/Security	Ratings, moderation, verification, reports, bans, audit logs
8 Admin	Admin dashboard, analytics, category/translation management
9 QA	Security, functional, mobile, localization, performance, abuse and edge-case testing
10 Pilot	Small local launch; measure fill rate, time-to-hire, completion and repeat use

23. Critical Business Rules

- A job cannot accept applications after FILLED, EXPIRED or CANCELLED.
- Selected workers cannot exceed required capacity.
- Only completed jobs generate normal rating opportunities.
- Only authorized roles can access role-specific dashboard/API operations.
- A single bad rating should not automatically ban an account; repeated or severe evidence triggers the trust process.

- Both workers and job providers are subject to the same core trust and safety standards.
- Historical jobs, ratings, disputes and audit records should be retained according to a defined retention policy.
- All important business rules are enforced server-side.

24. Launch Strategy

- Start in one district or cluster of villages rather than all India.
- Start with a focused set of high-demand categories.
- Launch with Marathi/Hindi/English if the pilot geography supports those languages.
- Recruit both sides of the marketplace: workers and job providers.
- Measure time-to-first-application, time-to-fill, completion rate, repeat usage and notification conversion.
- Expand categories, villages and languages only after local product-market fit is demonstrated.

25. Final Product Definition

Rural / Village Local Jobs is a multilingual, location-based two-sided marketplace SaaS connecting local workers with households, farmers, residents and local organizations. It combines worker skills, availability, proximity, reputation, job matching, notifications, chat/contact and automated job lifecycle management. The product has three separate dashboards — Worker, Job Provider and Admin — with strict role-based security. Both workers and providers are rated and can enter a trust enforcement process for repeated or severe bad behavior, including warnings, restrictions, suspension and bans. Jobs automatically become FILLED when the required number of workers is selected and EXPIRED after their application deadline, while historical records remain available for accountability and analytics.

26. Final Pre-Development Checklist

- Finalize first launch geography.
- Finalize first job categories and skill taxonomy.
- Finalize Marathi/Hindi/English translation set.
- Finalize worker/provider verification policy.
- Finalize rating and bad-rating enforcement thresholds.
- Finalize ban, suspension and appeal policy.
- Finalize job deadline defaults and cancellation rules.
- Finalize privacy/contact visibility rules.
- Finalize security threat model and access-control matrix.
- Finalize database ER diagram.
- Finalize API contracts.
- Finalize three-dashboard wireframes.
- Finalize notification rules and templates.
- Finalize MVP acceptance criteria.

Security note

Before production launch, obtain appropriate legal/privacy/security review for the target jurisdiction, especially for personal data, location, identity verification, communications, labor marketplace practices and future payments. This document is a product/engineering blueprint, not legal advice.